Limits Graphically and Numerically

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Estimating limits from a table

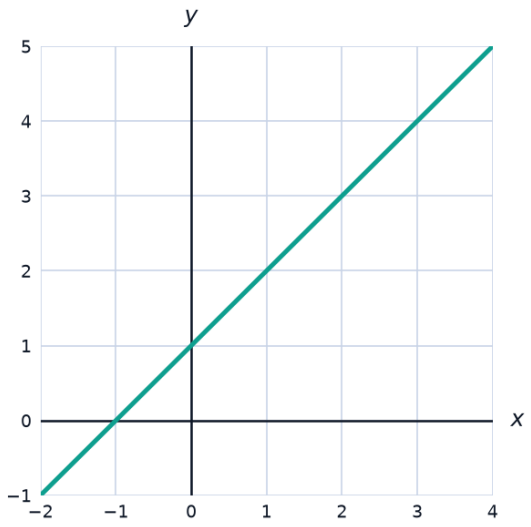
A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

2 [Foundational] • Target: Estimating limits from a table

Using a table of values with x near 0 (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

3 [Foundational] • Target: Reading limits from a graph

The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.



4 [Foundational] • Target: Reading limits from a graph

A function h has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether h is continuous at $x = 2$.

5 [Foundational] • Target: One-sided limits and limits that fail to exist

A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

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6 [Foundational] • Target: Algebraic evaluation

Evaluate each limit by direct substitution or factoring:

a $\lim_{x \to 4} (3x^2 - 5x + 1)$

b $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

c $\lim_{x \to -2} \frac{x^2 + 5x + 6}{x + 2}$

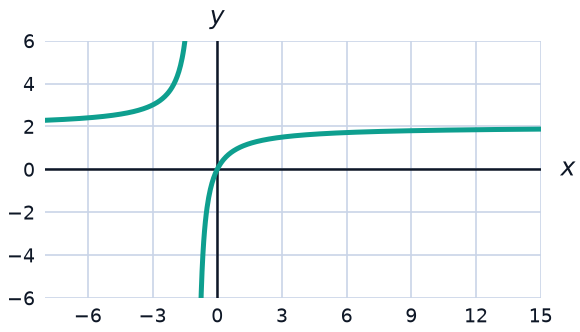
7 [Foundational] • Target: Algebraic evaluation

Evaluate $\lim_{x \to 0} \frac{\sqrt{x + 9} - 3}{x}$ by rationalizing the numerator.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits at infinity from a graph

The graph shows $f(x) = \frac{2x}{x + 1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x \to \infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x.



9 [Intermediate] • Target: Limits involving infinite discontinuities

For $f(x) = \frac{1}{x - 3}$, find $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.

10 [Intermediate] • Target: Limits involving infinite discontinuities

Evaluate $\lim_{x \to \infty} \frac{3x^2 + 1}{x^2 - 4}$, using the fact that the numerator and denominator have the same degree.

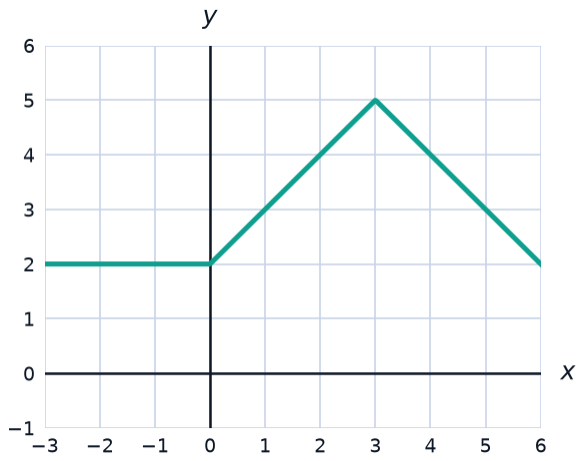
11 [Intermediate] • Target: Estimating limits with a calculator-style table

Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1,\ -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim_{x \to 0} \frac{e^x - 1}{x}$.

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12 [Intermediate] • Target: Limits with piecewise graphical behavior

A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim_{x \to 0} f(x)$ and $\lim_{x \to 3} f(x)$, stating whether each exists.



13 [Intermediate] • Target: Limits with piecewise graphical behavior

Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

14 [Intermediate] • Target: Limits involving infinite discontinuities

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Estimating limits with a calculator-style table • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Level 3 · AdvancedLet $f(x) = x^5 + 3x^3 + 2x - 1$. Find the equation of the tangent line to the graph of $y = f^{-1}(x)$ at $x = 5$.

16 [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 2.01 The Definition of the Derivative

Level 1 · FoundationsIf a student completes 24 of the 30 non-calculator questions in Part A with 10 minutes remaining, what pacing adjustment should they make?

17 [Challenging] • Target: Estimating limits from a table • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Level 2 · MediumDuring Section I Part B (Calculator), what are the four required calculator capabilities that College Board assesses?

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18 [Challenging] • Target: Reading limits from a graph • Cross-Lesson: 3.01 The Chain Rule

Level 2 · MediumThe graph of f' consists of a semicircle of radius 2 centered at the origin, lying above the x-axis. If $f(0) = 5$, find the value of $f(2)$.

19 [Challenging] • Target: One-sided limits and limits that fail to exist • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Level 3 · AdvancedLet $f(x) = x^3 - 3x - 1$. Explain how the Intermediate Value Theorem guarantees a root in $[1, 2]$.

20 [Challenging] • Target: Algebraic evaluation • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Level 1 · FoundationsEvaluate the limit $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Estimating limits from a table

Evaluate $\lim_{x \to 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

a. -3

b. 1

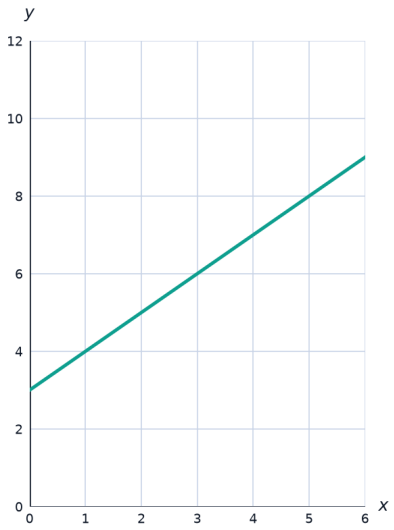
c. -1

d. 3

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22 [Foundational] • Target: Reading limits from a graph

Consider the function $f(x)$ defined piecewise as: $f(x) = \begin{cases} \frac{x^2-9}{x-3}, \text{ if } x \text{eq} 3; k, \text{ if } x = 3 \end{cases}$ For the function $f(x)$ to be continuous at $x = 3$, the value of k must be such that the limit of $f(x)$ as x approaches 3 exists and is equal to $f(3)$. Which of the following statements is true regarding the continuity of $f(x)$ at $x = 3$ if k is set to satisfy this condition, and what is the value of that limit?



a. The function is continuous at $x = 3$ and the limit is 6.

b. The function is continuous at $x = 3$ and the limit is 9.

c. The function is not continuous at $x = 3$ regardless of the value of k.

d. The function is continuous at $x = 3$ and the limit is 3.

23 [Foundational] • Target: One-sided limits and limits that fail to exist

The function $f(x)$ is defined piecewise as: $f(x) = \{ax^2 + 1, , x < 2; 5x - b, , x \geq 2$ If $f(x)$ is continuous at $x = 2$, and the limit of $f(x)$ as $x \to 2$ is 9, find the values of a and b.

a. a = 2, b = 1

b. a = 1, b = 1

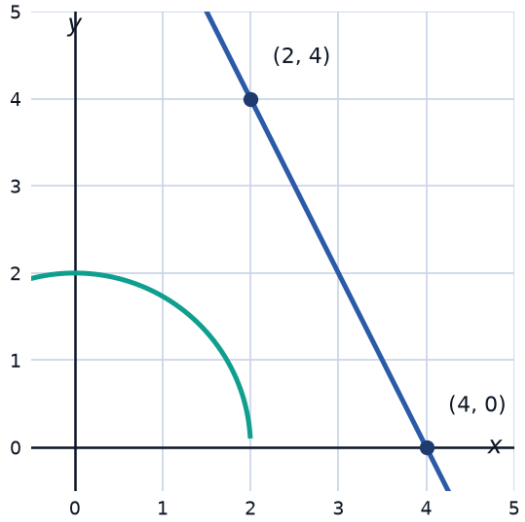
c. a = 2, b = -1

d. a = 1, b = -1

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24 [Foundational] • Target: Algebraic evaluation

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the t-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.



a. 4

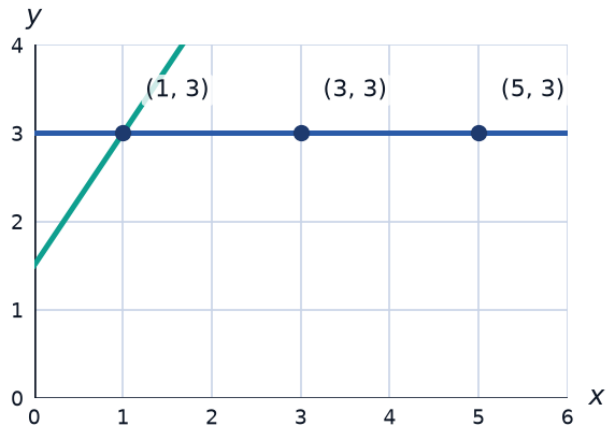
b. 6

c. 5

d. 7

25 [Foundational] • Target: Limits at infinity from a graph

The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.



a. 11

b. 14

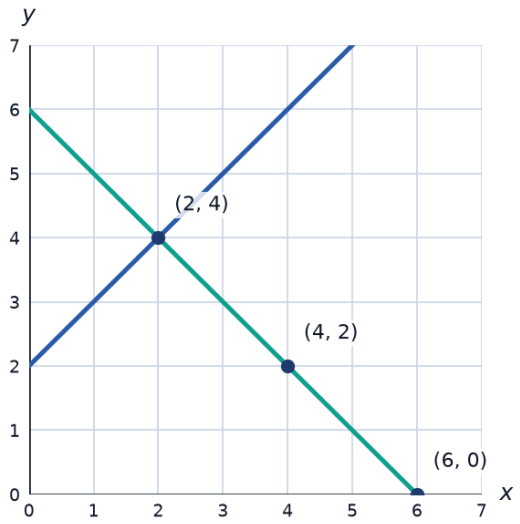
c. 17

d. 20

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26 [Foundational] • Target: Limits involving infinite discontinuities

The graph of the derivative of a function $f(x)$ is shown below. Given that $f(2) = 5$, determine the value of $f(6)$.



a. 13

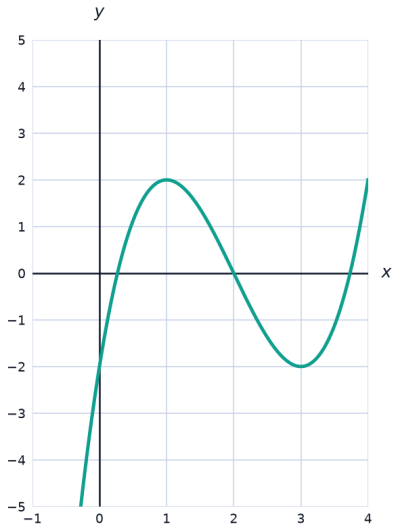
b. 18

c. 23

d. 28

27 [Foundational] • Target: Estimating limits with a calculator-style table

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.



a. -6

b. -12

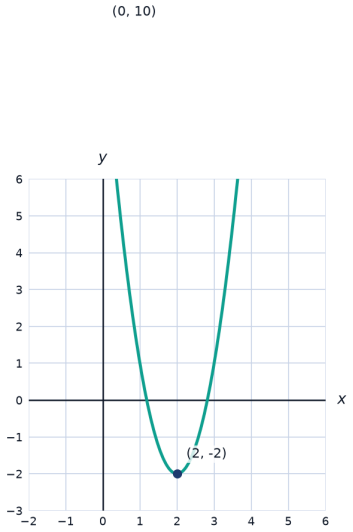
c. 0

d. 6

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28 [Foundational] • Target: Limits with piecewise graphical behavior

Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.



a. 15

b. 9

c. 11

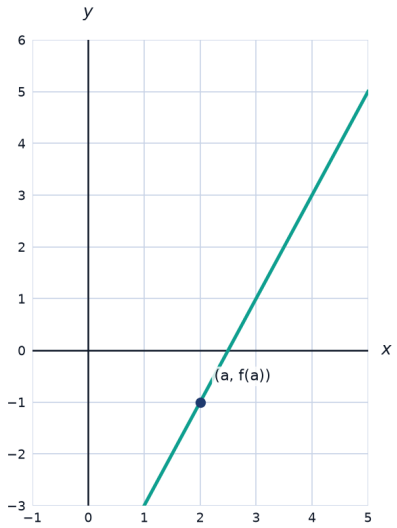
d. 21

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Estimating limits from a table**

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = a$ has the equation $y = 2x - 5$. If $f(a) = -1$ and $f'(a) = 2$, find the value of a.



a. -2

b. -1

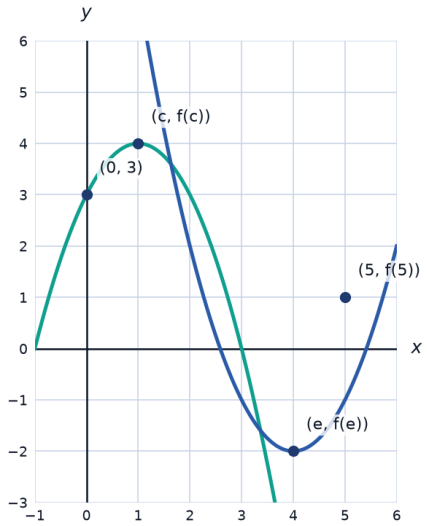
c. 1

d. 2

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30 **[Intermediate]** • **Target: Reading limits from a graph**

The graph of a differentiable function f is shown in the figure above. The function f has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of f, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. $*If*\int_{0}^{c} f(x) \, dx = k, *whatisthevalueof*\int_{c}^{5} f(x) \, dx*intermsof*k$?



a. 12 + k

b. 12 - k

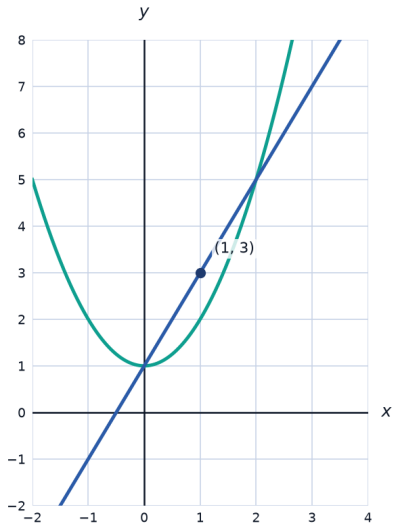
c. k - 12

d. k + 2

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31 **[Intermediate]** • **Target: One-sided limits and limits that fail to exist**

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.



a. 1

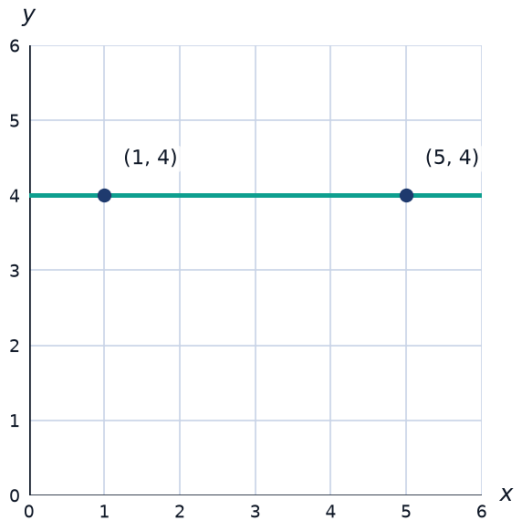
b. 2

c. 3

d. 4

32 **[Intermediate]** • **Target: Algebraic evaluation**

The graph of a function $f'(x)$ is shown below. It is given that $f(x)$ is a function such that $f(1) = 5$. The area between the curve $y = f'(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 5$ is represented by the shaded region. If the area of the shaded region is 16 square units, determine the value of $f(5)$.



a. 17

b. 21

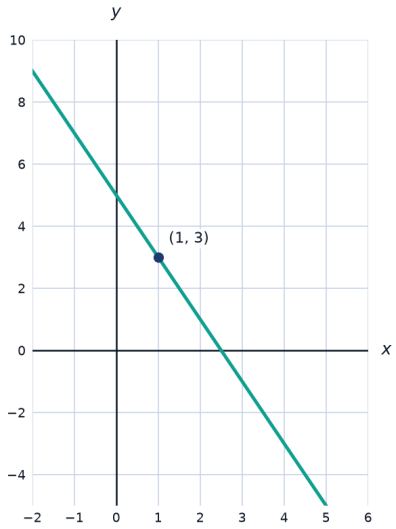
c. 9

d. 11

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33 **[Intermediate]** • **Target: Limits at infinity from a graph**

Consider the differential equation $\frac{dy}{dx} = kx - 3$, where k is a constant. The graph of a particular solution $y = f(x)$ to this differential equation is shown below. The tangent line to the graph at $x = 1$ has the equation $y = -2x + 5$. Find the value of k.



a. -3

b. -1

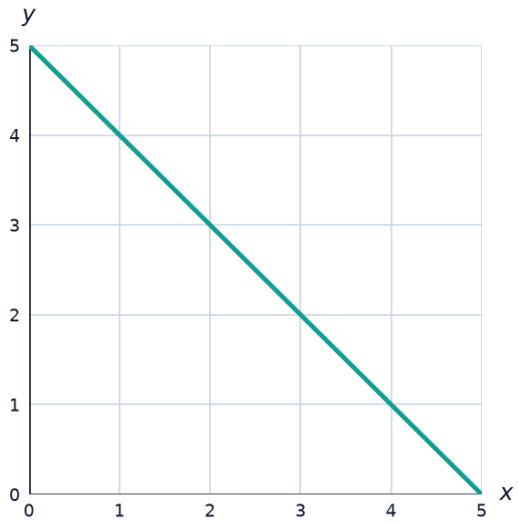
c. 1

d. 3

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34 **[Intermediate]** • **Target: Limits involving infinite discontinuities**

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?



a. 4.5

b. 6

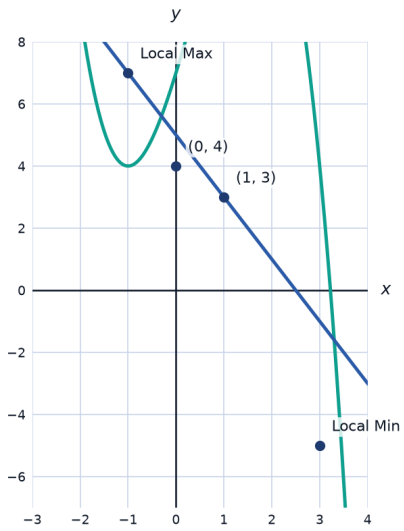
c. 7.5

d. 9

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35 **[Intermediate]** • **Target: Estimating limits with a calculator-style table**

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = -1$ and a local minimum at $x = 3$. The tangent line to the graph of f at $x = 1$ has the equation $y = -2x + 5$. If $f(0) = 4$ and $f(3) = -5$, find the value of $f'(1)$.



a. -5

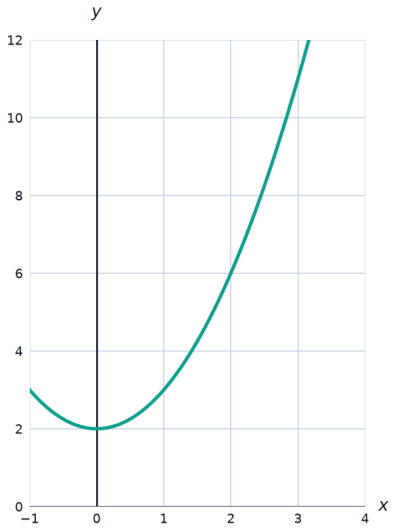
b. -2

c. 2

d. 5

36 **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

Consider the region bounded by the graph of $f(x) = x^2 + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 3$. Which of the following represents the area of this region?



a. $\int_0^3 (x^2 + 2)\,dx$

b. $\int_0^3 (x^2 - 2)\,dx$

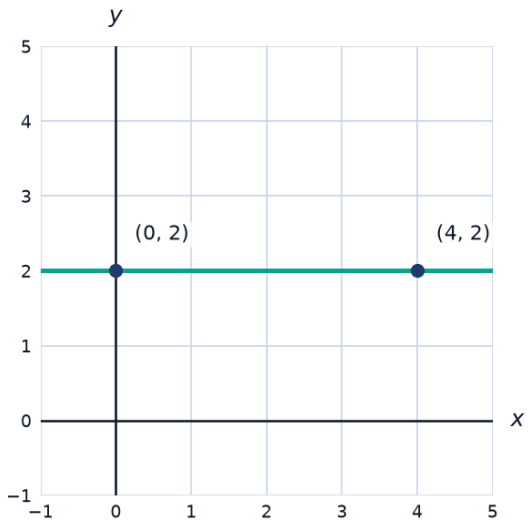
c. $\int_0^3 x^2\,dx$

d. $\int_0^3 2\,dx$

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37 **[Intermediate]** • **Target: Estimating limits from a table**

The graph of a function $f(x)$ is shown above. What is the instantaneous rate of change of $f(x)$ at $x = 2$?



a. -2

b. 0

c. 2

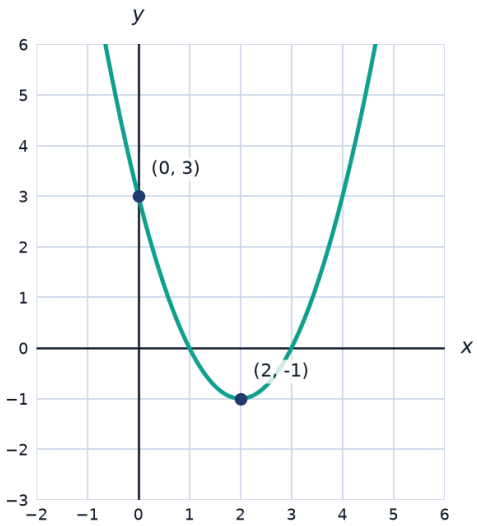
d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Reading limits from a graph • Cross-Lesson: 3.01 The Chain Rule

The graph of a function $f(x)$ is shown below. The function is a quadratic with a vertex at $(2, -1)$ and passes through the point $(0, 3)$. Which of the following statements correctly describes the derivative of $f(x)$ at $x = 2$?



a. The derivative is positive.

b. The derivative is negative.

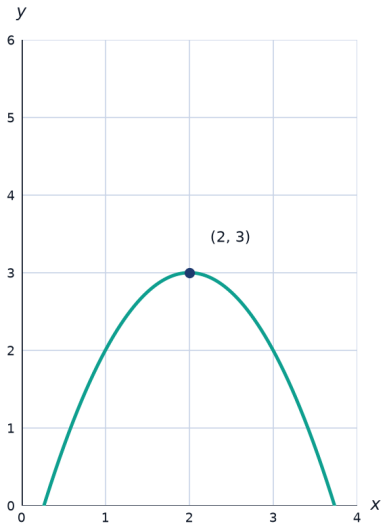
c. The derivative is zero.

d. The derivative is undefined.

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39 [Challenging] • Target: One-sided limits and limits that fail to exist • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.



a. 4

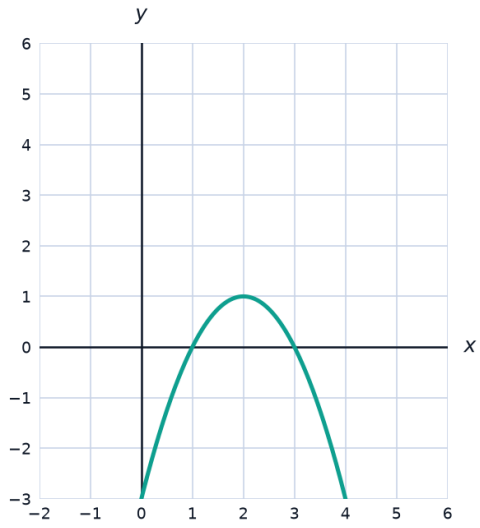
b. 5

c. 6

d. 7

40 [Challenging] • Target: Algebraic evaluation • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Which of the following represents the coordinates of the vertex of the parabola?



a. (2, 1)

b. (1, 0)

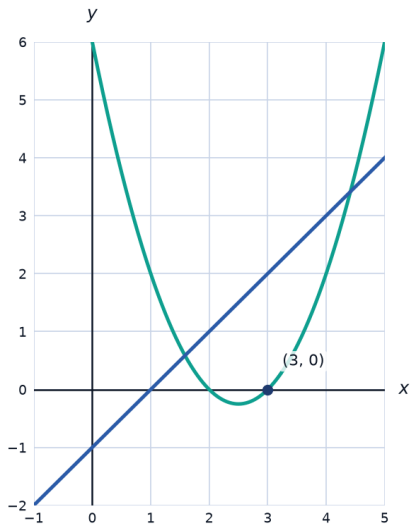
c. (3, 0)

d. (2, -1)

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41 **[Challenging]** • **Target: Limits at infinity from a graph** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Consider the function $f(x) = x^2 - 5x + 6$. What is the value of $\lim_{x \to 3} f(x)$?



a. 0

b. 3

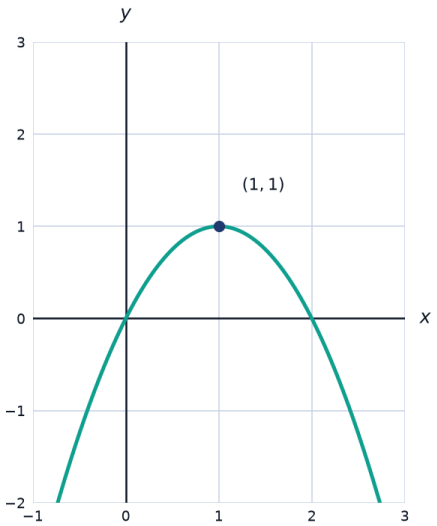
c. 6

d. 18

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42 **[Challenging]** • **Target: Limits involving infinite discontinuities** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

The graph of a differentiable function $f(x)$ is shown below. Which of the following statements about the derivative of $f(x)$, denoted $f'(x)$, at $x = 1$ is true?



a. $f'(1)$ is positive.

b. $f'(1)$ is negative.

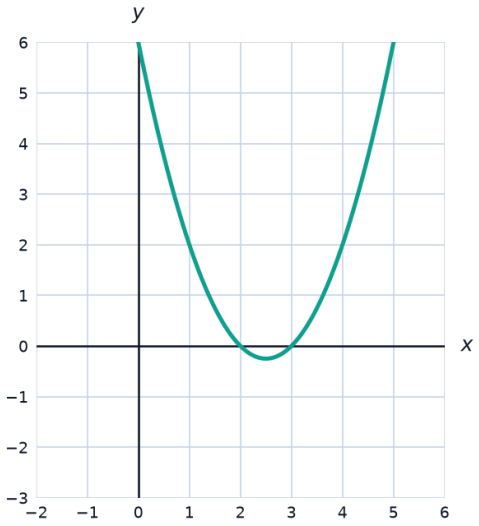
c. $f'(1)$ is zero.

d. $f'(1)$ cannot be determined from the graph.

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43 [Challenging] • Target: Estimating limits with a calculator-style table • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Consider the function $f(x) = x^2 - 5x + 6$. What is the limit of $f(x)$ as x approaches 2?



a. -1

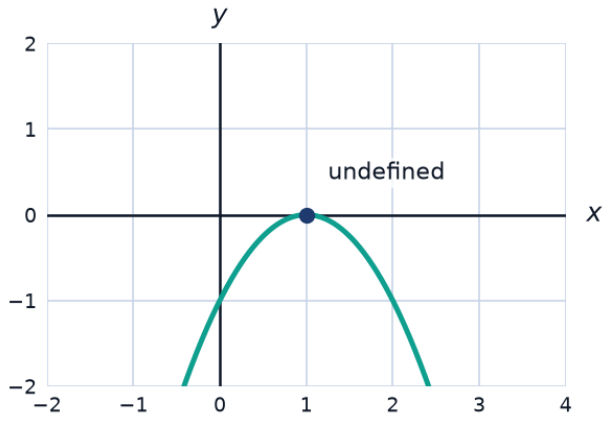
b. 0

c. 1

d. 2

44 [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Consider the function $f(x)$ graphed below. What is the limit of $f(x)$ as x approaches 1, i.e., $\lim_{x \to 1} f(x)$?



a. -1

b. 0

c. 1

d. 2

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45 [Challenging] • Target: Estimating limits from a table • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

A differentiable function f is defined on the interval $[-4, 4]$. The graph of f', the derivative of f, is shown in the figure above. The graph of f' consists of two line segments and a semicircle. (a) Find the value of $f(2) - f(-2)$. Show the setup for your answer. (b) For how many values of c in the interval $(-4, 4)$ does $f'(c) = 0$? Explain your reasoning. (c) The function f has a local maximum at $x = k$. Find the value of k. Explain your reasoning. (d) The function f has an inflection point at $x = p$. Find the value of p. Explain your reasoning.

a. There are no values of c for which $f'(c) = 0$.

b. There is one value of c for which $f'(c) = 0$.

c. There are two values of c for which $f'(c) = 0$.

d. There are three values of c for which $f'(c) = 0$.

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